

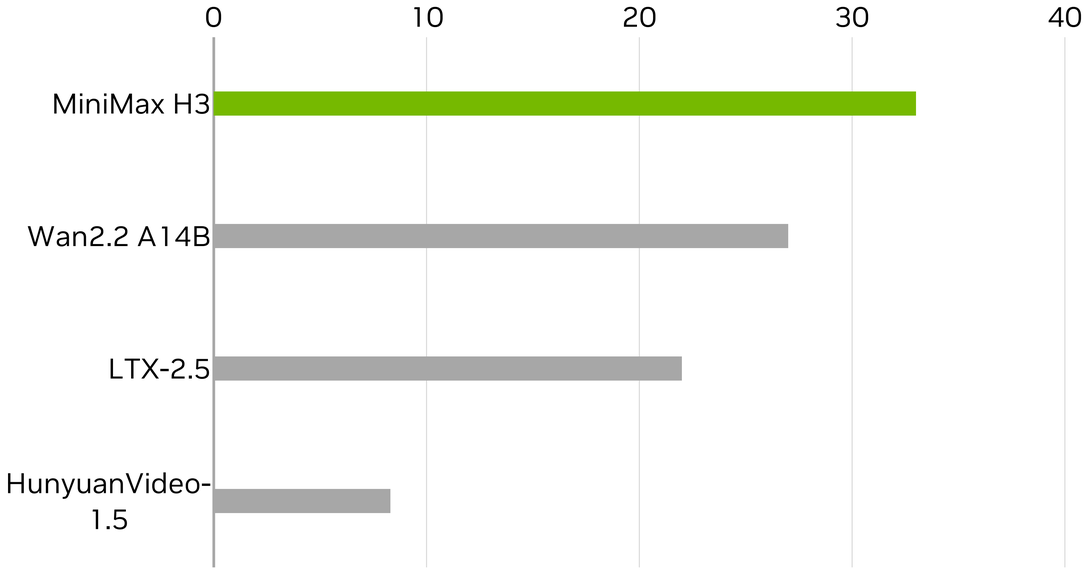
SM12x Inference Optimization with vLLM

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2026-08-23

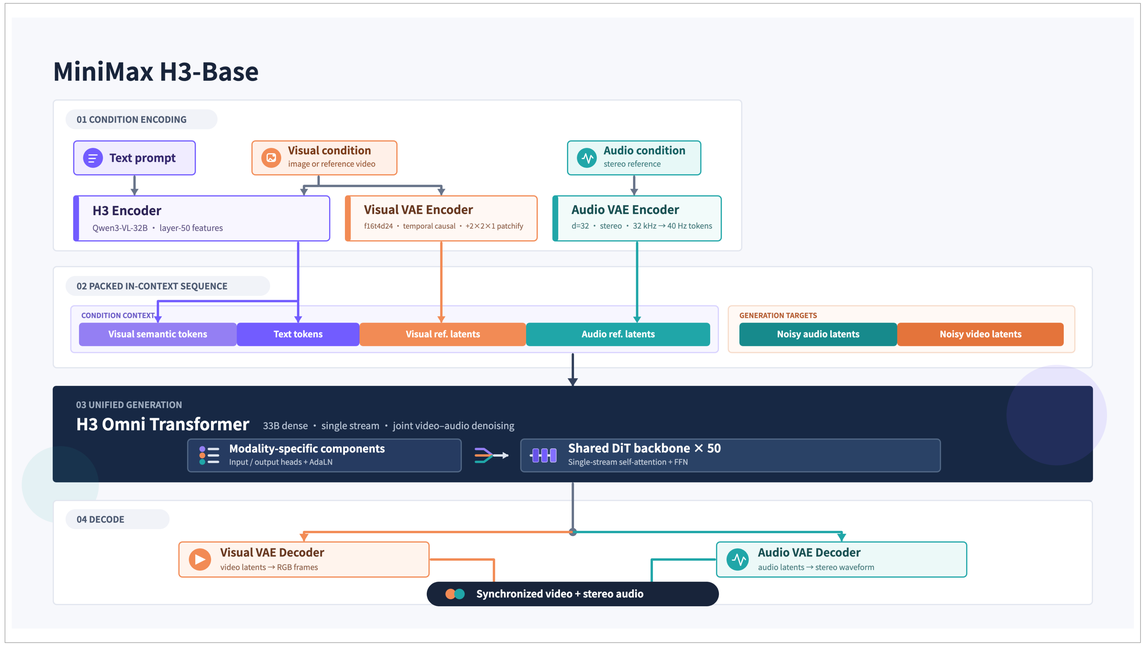
Real Time Is an End-to-End Budget

Generate about five seconds of audio and video within the same time budget.



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

The backbone is only part of the budget. Encoding, communication and audio–video output all contribute to latency.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

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SM120 Cards

GPU	SMs	Memory	Power	Interconnect
RTX PRO 5000	110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16 No GPU NVLink
RTX PRO 6000 **Workstation / Server**	188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	170	32 GB GDDR7	575 W	PCIe Gen5 ×16 No NVLink
DGX Spark **GB10 · SM121**	48	128 GB Unified LPDDR5X	140 W SoC	CPU–GPU NVLink-C2C 200 Gb/s ConnectX-7

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×8

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Baseline: Latency Breakdown

E2E: 213.402 s · Generated video: 960 × 544, 5.167 s

Reference configuration

- 8 × PRO 5000 GPUs (SM120)
- Original Ref2VA
- 50 steps · BF16 · dense attention
- DiT TP1 / Ulysses8
- Text encoder TP8 · VAE tiles8
- Native VAE · FP16 decode

GPU work by operator

	Mean s / GPU	Share
NCCL · includes waiting	122.182	**57.21%**
Attention	47.105	**22.06%**
GEMM	39.650	**18.57%**
Norm / activation / elementwise	2.288	1.07%
Layout / device copies	1.929	0.90%
CUDA memcpy / memset	0.064	0.03%
Convolution	0.192	0.09%
Other kernels	0.143	0.07%

Nsight GPU work includes overlap and waiting; shares exclude CPU work and idle gaps.

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Three Lessons for Real-Time Inference

Choose the optimization that changes the end-to-end bottleneck.

Reduce work with a quality budget

Fewer steps, sparse attention and lower precision reduce work.
Evaluate the quality tradeoff on the workload people will use.

Treat topology as part of the system

Communication can dominate GPU time.
Match data movement to the physical links before tuning kernels.

Follow the bottleneck as it moves

Decode, copies and output become more important as the model accelerates.
Re-profile after each major speedup; judge progress by end-to-end latency.

Read every speedup with its workload and configuration.

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Real Time Comes from the Whole Pipeline

960 × 544 · 5.167 s output · 8 GPUs · standalone H3

Stage	Cumulative change	Median (s)	E2E speedup vs. previous	RTF
00	Baseline · original BF16 / dense / 50 evaluations	213.402	—	41.304
01	+ PDD · 50 → 8 evaluations	44.327	4.814×†	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	1.710×	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	1.332×	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	2.848×	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	1.066×	1.240
06	+ Compute CUDA graphs	6.193	1.035×	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	1.143×	1.049
08–12	**+ Output and data-movement optimizations**	**4.922**	**1.101×**	**0.953**

08–12: fused tile / AAC preparation, direct I/O packing, compact routing and Q/K/V copy removal.

E2E speedup = previous displayed median ÷ current median. 08–12 compares the final result with stage 07.

Final median: 4.922 s · recheck: 4.927 s · 6/6 measured requests below 5 s

† Recorded median ratio; baseline refreshed separately (n = 2). Stages 01–12 retain n = 3 each; no isolated 50→8-step gain claimed.

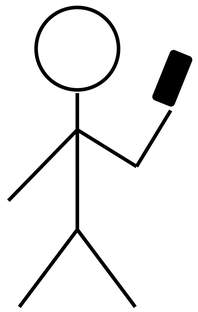
RTF = latency / 5.167 s. Standalone H3; ASR / LLM excluded. Model, schedule and precision changes are not all lossless.

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Real-Time Interaction Needs the Full Loop

Speech, reasoning and audio–video generation share the response budget.

“Hi, could you introduce yourself?”



User

Voice

Qwen3-ASR

0.6B · Speech → text

RECOGNIZED TEXT

“Hi, could you introduce yourself?”

Qwen3-4B

Reply → H3 prompt

ENGLISH PROMPT · EXCERPT

… “Hi, I’m Miles. I help with simple, clear answers.” … His lips follow the words.

Reply text + scene instructions

Reference image



H3

Audio + video

“Hi, I’m Miles…”



Avatar reply

Video + voice

Hear the avatar’s reply

960 × 544 · ~5 s per window · 8 × RTX PRO 5000

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Continuation Makes Clips Feel Continuous

The previous ending guides the next clip. Only new content is appended.

CHUNK A / LAST FRAME



CHUNK B / FIRST RETAINED FRAME



CONTINUOUS PLAYBACK

Trim B’s overlapping start

Chunk A · 5.167 s | Chunk B · +4.958 s

Recent audio–video context conditions the next generation; matching audio is trimmed with the overlap.

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Live Avatar: Recorded Conversation

74.55 seconds · user-recorded microphone conversation · native H3 audio + video



Play the recorded interaction · fullscreen avatar, live captions and timing overlays

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Takeaways

01 Multi-GPU optimization starts with topology

Map GPU/NIC links and locate communication bottlenecks.
Use available bandwidth efficiently, or reduce data transfer.
Communication is a key source of lossless speedups.

02 Evaluate quantization case by case

Lossy quantization behaves differently across models and workloads.
Measure quality and latency on representative cases.

03 Next: real-time response and richer interaction

Explore autoregressive diffusion for earlier responses.
Use prompt enhancers to support richer interaction.

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